



MOTORISED

BUTTERFLY VALVES

VALVES & DAMPERS



SLUICE & GATE VALVES



BALL VALVES



German Technology for Electrical Actuators



# MOTORISED VALVES & DAMPERS

## MOTORISED VALVES & DAMPERS

Marsh offers various types & sizes of Valves such as Butterfly, Ball, Sluice, Globe & Dampers with Electrical Actuators.

These units will be offered in "Ready for use" conditions with backup services of installation & commissioning assistance.

Marsh can also offer the allied electrical & electronic accessories such as control panels, feedback & positioning devices required in close-loop system.

### ***Electrically operated dampers***

We also undertake the assignments for complete designing & manufacturing of various types of dampers such as Motorised Butterfly type single flap dampers/ Multi louver dampers/ 3 way Diverter dampers /Material Diverters/Slide Gates/Guillotine dampers in IS 2062 , IS 2002, SA 516 , SS 316 , SS 304 , SS 310 etc. We also manufacture High Temperature & High Performance Dampers.

### ***Automated Valves & Dampers***

Electrically /Pneumatically Operated Butterfly Valves / Ball valves/ Slide Gates

Electrically Operated Globe Valves / Sluice / Gate Valves

Electrically Operated Guillotine & Other varieties of Dampers



BUTTERFLY & BALL VALVES

# MOTORISED / PNEUMATICALLY OPERATED BUTTERFLY & BALL VALVES

Marsh offers Motorised Butterfly Valves from size 50 to 1000 NB with line pressures of 6 to 16 Bar in wafer or flanged types, similarly Motorised Ball Valves from size 10 to 500 NB in the class of #150, #300, #600 & #800 end connections of screwed, socket weld or flanged types as detailed below :

## Special Features of Butterfly Valves

- Integrally moulded body liner ■ Longer Valve Life ■ Fit and forget valve ■ Low torque
- No need of separate gasket between the valve face & companion flange ■ Tight shut off ■ Easy Installation.

### DETAILS OF BUTTERFLY VALVES

| Valve Size | Type                                           | Material of Construction     |                         |                       |
|------------|------------------------------------------------|------------------------------|-------------------------|-----------------------|
|            |                                                | Body                         | Disc                    | Shaft                 |
| 50         | Wafer/<br>Flangeless<br>Or<br>Wafer/<br>Lugged | Cast Iron                    | Cast Iron               | Carbon Steel<br>S. S. |
| 65         |                                                |                              | SG Iron                 |                       |
| 80         |                                                |                              | encapsulated<br>in EPDM |                       |
| 100        |                                                | S. G. Iron                   |                         |                       |
| 125        |                                                | SG Iron Coated<br>with nylon |                         |                       |
| 150        | Double<br>Flanged<br>Body                      | Carbon Steel                 | S. S. (CF8)             |                       |
| 200        |                                                |                              | S. S. (CF8M)            |                       |
| 250        |                                                |                              | S. S.                   |                       |
| 300        |                                                | Aluminium bronze             |                         |                       |
| 350        |                                                |                              |                         |                       |
| 400        |                                                |                              |                         |                       |
| 450        |                                                |                              |                         |                       |
| 500        |                                                |                              |                         |                       |
| 600        |                                                |                              |                         |                       |
| 700        |                                                |                              |                         |                       |
| 800        |                                                |                              |                         |                       |
| 1000       |                                                |                              |                         |                       |



### TEST PRESSURE

| Seat  | Working Pressure | Test Pressures (Bar) |      |
|-------|------------------|----------------------|------|
|       |                  | Body                 | Seat |
| PN 10 | 10               | 15                   | 11   |
| PN 16 | 16               | 24                   | 17   |

\*For other material of Body, Ball, Disc contact Marsh Automation Pvt Ltd.

### DETAILS OF BALL VALVES

| VALVE<br>SIZE<br>MM | END<br>CONNECTIONS                   | TYPE             | PRESSURE<br>RATING | MATERIAL OF CONSTRUCTION |         |       |                    |
|---------------------|--------------------------------------|------------------|--------------------|--------------------------|---------|-------|--------------------|
|                     |                                      |                  |                    | BODY                     | BALL    | STEM  | SEAT               |
| 10                  | SCREWED<br>SOCKET<br>WELD<br>FLANGED | REGULAR          | CLASS              | CARBON                   | S.S.    | S. S. | PTFE /<br>G. F. T. |
| TO                  |                                      | FULL BORE        | # 150              | STEEL                    | 304/316 |       |                    |
| 500                 |                                      | ONE / TWO        | # 300              | FORGED                   | MIRROR  |       |                    |
|                     |                                      | THREE /<br>PIECE | # 600              | STEEL                    | FINISH  |       |                    |
|                     |                                      |                  | # 800              | S.S. 304/<br>316         |         |       |                    |



## Electrical Actuator details for Butterfly & Ball Valve Applications-Quarter Turn

Small Size Valve : These are direct mounted type compact actuators having following Features & accessories.

### Accessories

#### Standard

- Limit Switches (1NO +1NC) - 2 Nos. (Rating 10A - 250 VAC)
- Hand wheel
- Local position indicator
- Adjustable mechanical stoppers
- Space Heater

#### Optional

- Auxillary limit switches - (1NO +1NC) - 4 NOs.
- Torque limit switches (1NO +1NC) - 2 NOs. (for QT10 onwards)
- Three Position switch for ON/OFF operation Field/Panel Mounted
- Feed back Potentiometer - output 100-235 ohms or 0-1000 ohms
- Two wire position transmitter 4-20mA Output field/panel mounted
- Positioner is available in field/panel mounted to accept 4-20mA/0-10 V DC control signal & Rx Output.
- IP 67 Protection ■ Remote position indicator-Digital
- Special accessories on request ■ Local control station ■ Integral Starter

BUTTERFLY & BALL VALVES

| Actuator Data                            |  | Actuator Models             |      |       |       |       |       |       |
|------------------------------------------|--|-----------------------------|------|-------|-------|-------|-------|-------|
|                                          |  | QT-3                        | QT-5 | QT-10 | QT-20 | QT-30 | QT-40 | QT-50 |
| Rated Output Torque Kgm                  |  | 3                           | 5    | 10    | 20    | 30    | 40    | 50    |
| Operating Speed Sec/90°                  |  | 17                          | 17   | 20    | 20    | 26    | 26    | 26    |
| Switching Frequency<br>Operations / Hour |  |                             |      |       |       |       |       |       |
| a) Short Time                            |  |                             |      | 1200  |       |       |       |       |
| b) Normal                                |  |                             |      | 600   |       |       |       |       |
| Weight Kg.                               |  | 7.5                         | 9    | 12    | 15.5  | 37    | 37    | 37    |
| Degree of Protection                     |  | IP65/IP67                   |      |       |       |       |       |       |
| Area of Operation                        |  | Safe                        |      |       |       |       |       |       |
| <b>Motor Data</b>                        |  |                             |      |       |       |       |       |       |
| Type                                     |  | Non-Blocking A. C. Motor    |      |       |       |       |       |       |
| Supply : Standard                        |  | 220V AC 50Hz Single Phase   |      |       |       |       |       |       |
| Optional                                 |  | 110 V AC 50 Hz Single Phase |      |       |       |       |       |       |
|                                          |  | 415 VAC 50 Hz Three Phase   |      |       |       |       |       |       |
|                                          |  | 24 V DC                     |      |       |       |       |       |       |
| Output watts                             |  | 90                          | 125  | 175   | 250   | 290   | 300   | 320   |
| Nominal Current Amp                      |  | 0.5                         | 0.55 | 0.9   | 1.2   | 1.3   | 1.4   | 1.5   |
| Class Insulation                         |  | F                           |      |       |       |       |       |       |
| Ambient Temperature                      |  | 50° C                       |      |       |       |       |       |       |

Note :- For 24V DC & 415 VAC, 50 Hz motors above current & wattage values are not applicable

Large Size Valves:

These units are the combination of Rotary Actuators with Supplementary worm gear boxes having following features. :

Features of Emtork Actuators :

Standard Features

- Electric motor TEFC, 3ph, S-1 Duty
- Insulation class 'F'
- Travel Limits Switches - 2 NOs.(1 NO +1 NC )
- Torque Limits Switches - 2 NOs. (1 NO +1 NC )
- Continuous type local position indicator
- Handwheel with clutch mechanism
- Protection class IP 65 ■ Hammer-blow effect
- Output couplings viz A/B/C/D/E as per DIN 3210

Optional Features

- Electric motor - TESC, 3ph, S2 or S4 Duty
- Better class of insulation for motor
- Travel Limits Switches - 2 NO + 2 NC in place of 1 NO +1 NC - 2 NOs.
- Torque Limits Switches - 2 NO + 2 NC in place of 1 NO +1 NC - 2 NOs.
- Additional Travel limit switches ( 1 NO + 1 NC ) - 4 NOs. (Extra) - or ( 2 NO + 2 NC ) - 2 NOs., (4 NO + 4 NC) - 2 NOs
- Feedback potentiometer
- Remote position indicators-Analogue/Digital/Bar graph
- Space heater
- Electromagnetic, Electronic brake for the motor
- Protection class IP67/IP68
- Various types of local & remote panels
- Foot mounting brackets ■ Set of linkages
- Tailor made accessories to suit costumers' specific requirements
- Thermostats in motor winding
- Field / Panel mounted source to convert 220VAC to 24 VDC required for transmitter
- Supplementary Spur & Worm gear boxes
- Attachment for linear motion ■ **Retrofitting at site**



SCADA / DCS  
Compatible

MODBUS / PROFIBUS  
Protocol



| ACTUATOR DATA – LARGE SIZE BUTTERFLY VALVES |            |                         |                           |                              |
|---------------------------------------------|------------|-------------------------|---------------------------|------------------------------|
| Sr. No.                                     | Actuator   | Rated Output Torque kgm | Time of Operation Sec/90° | Remarks                      |
| 1                                           | M300/WG30  | 300                     | 3 to 45                   | Quarter Turn                 |
| 2                                           | M300/WG50  | 500                     | 5 to 75                   |                              |
| 3                                           | M300/WG75  | 750                     | 7.5 to 115                |                              |
| 4                                           | M300/WG100 | 1000                    | 10 to 150                 |                              |
| 5                                           | M500/WG75  | 1500                    | 7.5 to 115                | High Speed full Rotary Model |
| 6                                           | M500/WG100 | 2000                    | 10 to 150                 |                              |
| 7                                           | M500/WG31  | 500                     | 3 to 46                   |                              |
| 8                                           | M500/WG71  | 1500                    | 7.5 to 115                |                              |

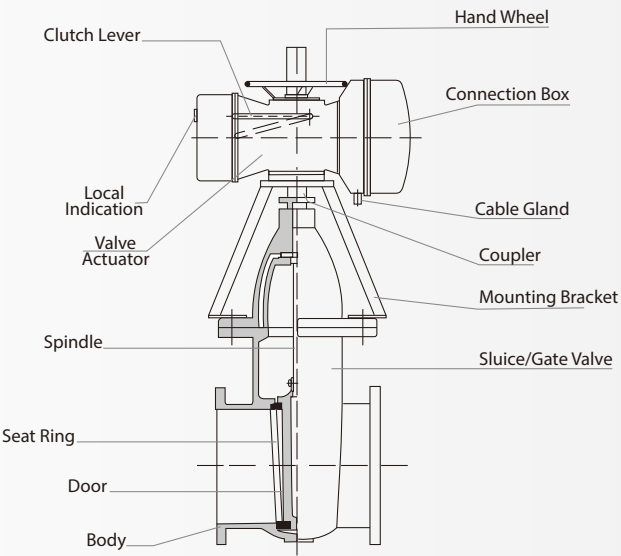
# MOTORISED SLUICE & GATE VALVES

MOTORISED SLUICE & GATE VALVES

We can offer various types & sizes of Sluice & Gate Valves duly coupled with Electrical Actuators & necessary automation accessories.

These Sluice valves generally confirm to IS 14846 and having flanges drilled as per IS 1538 Table 4 & 6 or can be customized in various pressure ratings such as PN 1.0, PN 1.6 etc. These valves are either rising or non rising spindle type. The Sluice valve body confirms to Cast Iron IS 210 Gr. FG 200/DI etc., The sizes are 50mm to 1000mm etc.

The Gate valves generally confirm to American standard in ASME Class 150, 300 etc. The Gate Valve body confirm to Cast Steel ASTM A 216 Gr. WCB/SS304/ SS316/F22 etc. The sizes are 50mm to 300mm. The forged steel Gate valves confirm to ASTM A105 are offered in the smaller range in Screwed, Socket Weld #800 & Flanged #150, #300 & #600. (Different types of material of construction can be offered ). The sizes are 15mm to 50mm. All the valves are Hydrostatic tested as per the pressure testing standard.



## Electrical Actuator details –Sluice / Gate & Knife Edge Gate Valves

The standard and optional features are same as given for large size butterfly valves. These are basically Rotary Actuators with various output Torques & Speeds and with different output couplings to suit rising & non rising stem Sluice & Gate Valves.

| TECHNICAL DATA FOR BASIC ACTUATORS WITH SUPPLIMENTARY SPUR GEAR BOXES |                |                              |                        |                                                                                                                      |
|-----------------------------------------------------------------------|----------------|------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------|
| Sr.No.                                                                | Actuator Model | Rated Output Torque Range Nm | Output Speed Range RPM | Remarks                                                                                                              |
| 1                                                                     | M30 To M200    | 30 To 200                    | 10 To 150              | Rotary Output Actuators with Various output Couplings viz A/B/C/D/E to Suit Rising & Non Rising Sluice & Gate Valves |
| 2                                                                     | M250 To M600   | 250 To 600                   | 10 To 150              |                                                                                                                      |
| 3                                                                     | M1000          | 1000                         | 10 To 120              |                                                                                                                      |
| 4                                                                     | M300 / SG08    | 2000                         | 1.25 To 19             |                                                                                                                      |
| 5                                                                     | M300 / SG12    | 3000                         | 0.8 To 12.5            |                                                                                                                      |
| 6                                                                     | M500 / SG212   | 5000                         | 0.8 To 12.5            |                                                                                                                      |
|                                                                       |                |                              |                        |                                                                                                                      |

# MOTORISED GLOBE VALVES FOR ON/OFF & CONTROL APPLICATION

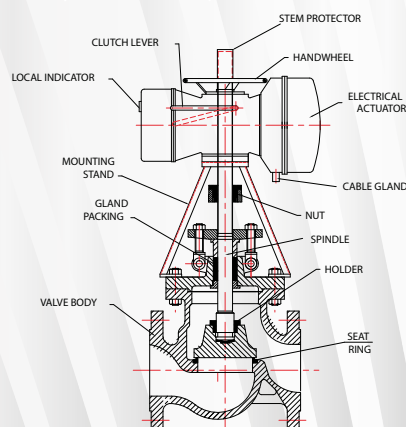
## DETAILS OF GLOBE VALVES

| Sr. No. | Valve Size mm | Material of Construction                                     | Pressure Class | End Connection                              |
|---------|---------------|--------------------------------------------------------------|----------------|---------------------------------------------|
| 1       | 15            | Cast Iron / SGI / DI / Cast Carbon Steel / Forged Steel / SS | 150            | Screwed / Socket Weld / Butt Weld / Flanged |
| 2       | 20            |                                                              |                |                                             |
| 3       | 25            |                                                              |                |                                             |
| 4       | 40            |                                                              | 300            |                                             |
| 5       | 50            |                                                              | 600            |                                             |
| 6       | 65            |                                                              |                |                                             |
| 7       | 80            |                                                              | 800            |                                             |
| 8       | 100           |                                                              |                |                                             |
| 9       | 125           |                                                              | 900            |                                             |
| 10      | 150           |                                                              |                |                                             |
| 11      | 200           |                                                              | 1500           |                                             |
| 12      | 250           |                                                              |                |                                             |
| 13      | 300           |                                                              | 2500           |                                             |

Globe Valves in various sizes & pressure ratings suitable for On/Off and control applications in the process Industry can be offered duly mounted with Electrical Actuators, depending upon the thrust requirements the actuators can be offered in Single or Three Phase Electric supply. The accessories such as Positioners, Feedback Transmitters, which are required for close loop systems can also be offered as a complete system.

## DETAILS OF ELECTRICAL ACTUATORS FOR GLOBE VALVES

| Sr. No. | Actuator Model              | Stroke Length mm | Operating Speed mm/Sec | Rated Actuating Force kgs. | Electric Supply                                                             |
|---------|-----------------------------|------------------|------------------------|----------------------------|-----------------------------------------------------------------------------|
| 1       | ST-5102-32                  | 40               | 0.26                   | 200                        | 220 VAC<br>50 Hz 1 Phase<br>or<br>110 VAC<br>50 Hz 1 Phase<br>or<br>24 V DC |
| 2       | ST-5102-33                  | 40/60            | 0.52                   | 200                        |                                                                             |
| 3       | ST-5103-61                  | 60               | 0.45                   | 600                        |                                                                             |
| 4       | ST-5103-76                  | 60/75            | 0.8                    | 600                        |                                                                             |
| 5       | M60 with Linear Attachment  | 100              | 0.8 to 12.5            | 2000                       | 415 VAC<br>50 Hz 3 Phase                                                    |
| 6       | M150 with Linear Attachment | 120              | 1.0 to 15              | 3500                       | 380 VAC<br>60 Hz 3 Phase<br><br>480 VAC<br>60 Hz 3 Phase                    |
| 7       | M250 with Linear Attachment | 160              | 1.16 to 17.5           | 6500                       |                                                                             |
| 8       | M500 with Linear Attachment | 160              | 1.33 to 20             | 10000                      |                                                                             |



MOTORISED GLOBE VALVE